CHAPTER 10 - Hypothesis testing: Two samples

SDS188

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1 Hypothesis testing for the difference in population means - independent samples

Is there a significant difference between population means?

Assumptions:

- Populations are not related
- Samples are collected randomly and independently
- Populations are normally distributed

1.1 Population standard deviations are known

σ_1 and σ_2 are known $\longrightarrow$ Z-distribution

1.1.1 Confidence interval

$$(1-\alpha)\%\text{CI for } \mu_1-\mu_2 : (\bar{x}_1-\bar{x}_2) \pm z_{\frac{\alpha}{2}}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}$$

Conclusion: Is 0 within the interval?

0 is in the interval	Do not reject the H_0
0 is in the interval	Reject the H_0

1.1.2 Hypothesis tests

Critical value approach

	Left-sided test	**Two-sided test**	**Right-sided test**
1	H_0: $\mu_1 - \mu_2 = 0$ (no difference) H_1: $\mu_1 - \mu_2 < 0$ ($\mu_1 < \mu_2$)	H_0: $\mu_1 - \mu_2 = 0$ (no difference) H_1: $\mu_1 - \mu_2 \neq 0$ ($\mu_1 \neq \mu_2$)	H_0: $\mu_1 - \mu_2 = 0$ (no difference) H_1: $\mu_1 - \mu_2 > 0$ ($\mu_1 > \mu_2$)
2	α, n_1, n_2		
3	σ_1 and σ_2 are known $\longrightarrow$ Z-test		
4	z_α	$z_{\frac{\alpha}{2}}$ and $z_{1-\frac{\alpha}{2}}$	$z_{1-\alpha}$
5	$z_{stat} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$		
6	Reject H_0 if $z_{stat} < z_\alpha$, do not reject H_0 if $z_{stat} \geq z_\alpha$	Reject H_0 if $z_{stat} < z_{\frac{\alpha}{2}}$ or if $z_{stat} > z_{1-\frac{\alpha}{2}}$, do not reject H_0 if $z_{\frac{\alpha}{2}} \leq z_{stat} \leq z_{1-\frac{\alpha}{2}}$	Reject H_0 if $z_{stat} > z_{1-\alpha}$, do not reject H_0 if $z_{stat} \leq z_{1-\alpha}$
7	Reject H_0: Since $z_{stat} < z_\alpha$, reject the H_0. There is sufficient evidence that μ_1 is smaller than μ_2	Reject H_0: Since $z_{stat} < z_{\frac{\alpha}{2}}$ (or $z_{stat} > z_{1-\frac{\alpha}{2}}$), reject H_0. There is sufficient evidence that μ_1 differs from μ_2	Reject H_0: Since $z_{stat} > z_{1-\alpha}$, reject the H_0. There is sufficient evidence that μ_1 is larger than μ_2

	Do not reject H_0: Since $z_{stat} \geq z_{\alpha}$, do not reject the H_0. There is not sufficient evidence of a difference between μ_1 and μ_2 or that μ_1 is smaller than μ_2	Do not reject H_0: Since $z_{\frac{\alpha}{2}} \leq z_{stat} \leq z_{1-\frac{\alpha}{2}}$, do not reject the H_0. There is not sufficient evidence that μ_1 differs from μ_2	Do not reject H_0: Since $z_{stat} \leq z_{1-\alpha}$, do not reject the H_0. There is not sufficient evidence of a difference between μ_1 and μ_2 or that μ_1 is larger than μ_2

p-value approach

	Left-sided test	**Two-sided test**	**Right-sided test**
1	H_0: $\mu_1 - \mu_2 = 0$ (no difference) H_1: $\mu_1 - \mu_2 < 0$ ($\mu_1 < \mu_2$)	H_0: $\mu_1 - \mu_2 = 0$ (no difference) H_1: $\mu_1 - \mu_2 \neq 0$ ($\mu_1 \neq \mu_2$)	H_0: $\mu_1 - \mu_2 = 0$ (no difference) H_1: $\mu_1 - \mu_2 > 0$ ($\mu_1 > \mu_2$)
2	α, n_1, n_2		
3	σ_1 and σ_2 are known $\longrightarrow$ Z-test		
4	$z_{stat} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$		
5	p-value = $P(Z < z_{stat})$	If $z_{stat} < 0$, p-value = $2P(Z < z_{stat})$ or if $z_{stat} > 0$, p-value = $2P(Z > z_{stat})$	p-value = $P(Z > z_{stat})$
6	Reject H_0: Since p-value $< \alpha$, reject the H_0. There is sufficient evidence that μ_1 is smaller than μ_2	Reject H_0: Since p-value $< \alpha$, reject the H_0. There is sufficient evidence that μ_1 differs from μ_2	Reject H_0: Since p-value $< \alpha$, reject the H_0. There is sufficient evidence that μ_1 is larger than μ_2

	Do not reject H_0: Since p-value $\geq \alpha$, do not reject the H_0. There is not sufficient evidence of a difference between μ_1 and μ_2 or that μ_1 is smaller than μ_2	Do not reject H_0: Since p-value $\geq \alpha$, do not reject the H_0. There is not sufficient evidence of a difference between μ_1 and μ_2	Do not reject H_0: Since p-value $\geq \alpha$, do not reject the H_0. There is not sufficient evidence of a difference between μ_1 and μ_2 or that μ_1 is larger than μ_2

1.2 Population standard deviations are unknown

$\sigma_1 \approx \sigma_2$ (there should at least be an assumption that the variances are equal)

σ_1 and σ_2 are unknown $\longrightarrow$ t-test

1.2.1 Confidence interval

$$(1-\alpha)\%\text{CI for } \mu_1 - \mu_2 : (\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2},\text{df}} \sqrt{s_p^2 (\frac{1}{n_1} + \frac{1}{n_2})}$$

where df $= n_1 + n_2 - 2$ en

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 - 1) + (n_2 - 1)}$$

Conclusion: Is 0 within the interval?

0 is in the interval	Do not reject the H_0
0 is in the interval	Reject the H_0

1.2.2 Hypothesis tests

Critical value approach

	Left-sided test	**Two-sided test**	**Right-sided test**
1	H_0: $\mu_1 - \mu_2 = 0$ (no difference) H_1: $\mu_1 - \mu_2 < 0$ ($\mu_1 < \mu_2$)	H_0: $\mu_1 - \mu_2 = 0$ (no difference) H_1: $\mu_1 - \mu_2 \neq 0$ ($\mu_1 \neq \mu_2$)	H_0: $\mu_1 - \mu_2 = 0$ (no difference) H_1: $\mu_1 - \mu_2 > 0$ ($\mu_1 > \mu_2$)
2	α, n_1, n_2		
3	σ_1 and σ_2 are unknown $\longrightarrow$ t-test		
4	$t_{\alpha,(n_1+n_2-2)}$	$t_{\frac{\alpha}{2},(n_1+n_2-2)}$ and $t_{1-\frac{\alpha}{2},(n_1+n_2-2)}$	$t_{1-\alpha,(n_1+n_2-2)}$
5	$t_{stat} = \frac{(\bar{x}_1-\bar{x}_2)-0}{\sqrt{s_p^2(\frac{1}{n_1}+\frac{1}{n_2})}}$		
6	Reject H_0 if $t_{stat} < t_{\alpha,(n_1+n_2-2)}$, do not reject H_0 as $t_{stat} \geq t_{\alpha,(n_1+n_2-2)}$	Reject H_0 as $t_{stat} < t_{\frac{\alpha}{2},(n_1+n_2-2)}$ or if $t_{stat} > t_{1-\frac{\alpha}{2},(n_1+n_2-2)}$, do not reject H_0 as $t_{\frac{\alpha}{2},(n_1+n_2-2)} \leq t_{stat} \leq t_{1-\frac{\alpha}{2},(n_1+n_2-2)}$	Reject H_0 as $t_{stat} > t_{1-\alpha,(n_1+n_2-2)}$, do not reject H_0 as $t_{stat} \leq t_{1-\alpha,(n_1+n_2-2)}$
7	Reject H_0: Since $t_{stat} < t_{\alpha,(n_1+n_2-2)}$, reject the H_0. There is sufficient evidence that μ_1 is smaller than μ_2	Reject H_0: Since $t_{stat} < t_{\frac{\alpha}{2},(n_1+n_2-2)}$ (or $t_{stat} > t_{1-\frac{\alpha}{2},(n_1+n_2-2)}$), reject the H_0. There is sufficient evidence of a difference between μ_1 and μ_2	Reject H_0: Since $t_{stat} > t_{1-\alpha,(n_1+n_2-2)}$, reject the H_0. There is sufficient evidence that μ_1 is larger than μ_2

	Do not reject H_0: Since $t_{stat} \geq t_{\alpha,(n_1+n_2-2)}$, do not reject the H_0. There is not sufficient evidence that there is a difference between μ_1 and μ_2 or that μ_1 is smaller than μ_2	Do not reject H_0: Since $t_{\frac{\alpha}{2},(n_1+n_2-2)} \leq t_{stat} \leq t_{1-\frac{\alpha}{2},(n_1+n_2-2)}$, do not reject the H_0. There is not sufficient evidence that μ_1 differs from μ_2	Do not reject H_0: Since $t_{stat} \leq t_{1-\alpha,(n_1+n_2-2)}$, do not reject the H_0. There is not sufficient evidence that there is a difference between μ_1 and μ_2 or that μ_1 is larger than μ_2

p-value approach

	Left-sided test	**Two-sided test**	**Right-sided test**
1	H_0: $\mu_1 - \mu_2 = 0$ (no difference) H_1: $\mu_1 - \mu_2 < 0$ ($\mu_1 < \mu_2$)	H_0: $\mu_1 - \mu_2 = 0$ (no difference) H_1: $\mu_1 - \mu_2 \neq 0$ ($\mu_1 \neq \mu_2$)	H_0: $\mu_1 - \mu_2 = 0$ (no difference) H_1: $\mu_1 - \mu_2 > 0$ ($\mu_1 > \mu_2$)
2	α, n_1, n_2		
3	σ_1 and σ_2 are unknown $\longrightarrow$ t-test		
4	$t_{stat} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{s_p^2(\frac{1}{n_1} + \frac{1}{n_2})}}$		
5	p-value = $P(t < t_{stat})$, df $= n_1 + n_2 - 2$	If $t_{stat} < 0$, p-value = $2P(t < t_{stat})$ or if $t_{stat} > 0$, p-value = $2P(t > t_{stat})$, df $= n_1 + n_2 - 2$	p-value = $P(t > t_{stat})$, df $= n_1 + n_2 - 2$

6	Reject H_0: Since p-value $<$ α, reject the H_0. There is sufficient evidence that μ_1 is smaller than μ_2 Do not reject H_0: Since p-value $\geq$ α, do not reject the H_0. There is not sufficient evidence that there is a difference between μ_1 and μ_2 or that μ_1 is smaller than μ_2	Reject H_0: Since p-value $<$ α, reject the H_0. There is sufficient evidence of a difference between μ_1 and μ_2 Do not reject H_0: Since p-value $\geq$ α, do not reject the H_0. There is not sufficient evidence that μ_1 differs from μ_2	Reject H_0: Since p-value $<$ α, reject the H_0.There is sufficient evidence that μ_1 is larger than μ_2 Do not reject H_0: Since p-value $\geq$ α, do not reject the H_0. There is not sufficient evidence that there is a difference between μ_1 and μ_2 or that μ_1 is larger than μ_2

2 Hypothesis testing for the difference in population means - related samples

Paired differences

Paired variables X_1 and X_2. Let d_i be the difference between paired variables. The number of observations for each variable is the same: $n_1 = n_2 = n$

	$\mathbf{x}_{i1}$	$\mathbf{x}_{i2}$	$\mathbf{d}_i$
1	x_{11}	x_{12}	$d_1 = x_{11} - x_{12}$
2	x_{21}	x_{22}	$d_2 = x_{21} - x_{22}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	x_{n1}	x_{n2}	$d_n = x_{n1} - x_{n2}$

Average of paired differences:

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i$$

Standard deviation of paired differences:

$$s_d = \frac{1}{n-1}\sum_{i=1}^{n}(d_i - \bar{d})^2$$

Standard error of the sample mean of paired differences:

$$\sigma_{\bar{d}} = \frac{s_d}{\sqrt{n}}$$

Assumptions:

Differences between paired values are normally distributed ($d_i \sim N(\mu_d, \frac{\sigma_d^2}{n})$)

Test statistic:

$$t_{stat} = \frac{\bar{d} - \mu_d}{\frac{s_d}{\sqrt{n}}}$$

Degrees of freedom: $df = n - 1$

2.1 Confidence interval

$$(1-\alpha)\%\text{CI for } \mu_d : \bar{d} \pm t_{\frac{\alpha}{2},\text{df}}\frac{s_d}{\sqrt{n}}$$

Conclusion: Is 0 within the interval?

0 is in the interval	Do not reject the H_0
0 is in the interval	Reject the H_0

2.2 Hypothesis tests

2.2.1 Critical value approach

	Left-sided test	**Two-sided test**	**Right-sided test**
1	H_0: $\mu_d = 0$ (no difference) H_1: $\mu_d < 0$	H_0: $\mu_d = 0$ (no difference) H_1: $\mu_d \neq 0$	H_0: $\mu_d = 0$ (no difference) H_1: $\mu_d > 0$
2	α and n		
3	σ is unknown $\longrightarrow$ t-test		
4	$t_{\alpha,(n-1)}$	$t_{\frac{\alpha}{2},(n-1)}$ en $t_{1-\frac{\alpha}{2},(n-1)}$	$t_{1-\alpha,(n-1)}$
5	$t_{stat} = \frac{\bar{d}-0}{\frac{s_d}{\sqrt{n}}}$		
6	Reject H_0 if $t_{stat} < t_{\alpha,(n-1)}$, do not reject H_0 as $t_{stat} \geq t_{\alpha,(n-1)}$	Reject H_0 if $t_{stat} < t_{\frac{\alpha}{2},(n-1)}$ or if $t_{stat} > t_{1-\frac{\alpha}{2},(n-1)}$, do not reject H_0 if $t_{\frac{\alpha}{2},(n-1)} \leq t_{stat} \leq t_{1-\frac{\alpha}{2},(n-1)}$	Reject H_0 if $t_{stat} > t_{1-\alpha,(n-1)}$, do not reject H_0 if $t_{stat} \leq t_{1-\alpha,(n-1)}$
7	Reject H_0: Since $t_{stat} < t_{\alpha,(n-1)}$, reject the H_0. There is sufficient evidence that μ_d is smaller than 0 Do not reject H_0: Since $t_{stat} \geq t_{\alpha,(n-1)}$, do not reject the H_0. There is not sufficient evidence that there is a difference between the means.	Reject H_0: Since $t_{stat} < t_{\frac{\alpha}{2},(n-1)}$ (of $t_{stat} > t_{1-\frac{\alpha}{2},(n-1)}$), reject the H_0. There is sufficient evidence that μ_d differs from 0 Do not reject H_0: Since $t_{\frac{\alpha}{2},(n-1)} \leq t_{stat} \leq t_{1-\frac{\alpha}{2},(n-1)}$, do not reject the H_0. There is not sufficient evidence that there is a difference between the means.	Reject H_0: Since $t_{stat} > t_{1-\alpha,(n-1)}$, reject the H_0. There is sufficient evidence that μ_d is larger than 0 Do not reject H_0: Since $t_{stat} \leq t_{1-\alpha,(n-1)}$, do not reject the H_0. There is not sufficient evidence that there is a difference between the means.

2.2.2 p-value approach

	Left-sided test	**Two-sided test**	**Right-sided test**
1	H_0: $\mu_d = 0$ (no difference) H_1: $\mu_d < 0$	H_0: $\mu_d = 0$ (no difference) H_1: $\mu_d \neq 0$	H_0: $\mu_d = 0$ (no difference) H_1: $\mu_d > 0$
2	α and n		
3	σ is unknown $\longrightarrow$ t-test		
4	$t_{stat} = \frac{\bar{d}-0}{\frac{s_d}{\sqrt{n}}}$		
5	p-value = $P(t < t_{stat})$, df = $n-1$	If $t_{stat} < 0$, p-value = $2P(t < t_{stat})$ or if $t_{stat} > 0$, p-value = $2P(t > t_{stat})$, df = $n-1$	p-value = $P(t > t_{stat})$, df = $n-1$
6	Reject H_0: Since p-value $< \alpha$, reject the H_0. There is sufficient evidence that μ_d is smaller than 0 Do not reject H_0: Since p-value $\geq \alpha$, do not reject the H_0. There is not sufficient evidence There is not sufficient evidence that there is a difference between the means.	Reject H_0: Since p-value $< \alpha$, reject the H_0. There is sufficient evidence that μ_d differs from 0 Do not reject H_0: Since p-value $\geq \alpha$, do not reject the H_0. There is not sufficient evidence There is not sufficient evidence that there is a difference between the means.	Reject H_0: Since p-value $< \alpha$, reject the H_0. There is sufficient evidence that μ_d is larger than 0 Do not reject H_0: Since p-value $\geq \alpha$, do not reject the H_0. There is not sufficient evidence that there is a difference between the means.

3 Hypothesis testing for the difference in population proportions

Test for the difference between two population proportions $\pi_1 - \pi_2$

Assumptions:

If

$n_1\pi_1 \geq 5$ and $n_1(1-\pi_1) \geq 5$

$n_2\pi_2 \geq 5$ and $n_2(1-\pi_2) \geq 5$

$(p_1 - p2)$ is normally distributed

Let p_1 and p_2 be estimates for π_1 and π_2

3.1 Confidence interval

$$(1-\alpha)\%\text{CI for } \pi_1 - \pi_2 : (p_1 - p_2) \pm z_{\frac{\alpha}{2}} \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

Conclusion: Is 0 within the interval?

0 is in the interval	Do not reject the H_0
0 is in the interval	Reject the H_0

3.2 Hypothesis tests

3.2.1 Critical value approach

	Left-sided test	**Two-sided test**	**Right-sided test**
1	H_0: $\pi_1 - \pi_2 = 0$ (no difference) H_1: $\pi_1 - \pi_2 < 0$ ($\pi_1 < \pi_2$)	H_0: $\pi_1 - \pi_2 = 0$ (no difference) H_1: $\pi_1 - \pi_2 \neq 0$ ($\pi_1 \neq \pi_2$)	H_0: $\pi_1 - \pi_2 = 0$ (no difference) H_1: $\pi_1 - \pi_2 > 0$ ($\pi_1 > \pi_2$)
2	α, n_1, n_2		

3	z_α	$z_{\frac{\alpha}{2}}$ and $z_{1-\frac{\alpha}{2}}$	$z_{1-\alpha}$
4	$\bar{p} = \frac{x_1+x_2}{n_1+n_2}$ $z_{stat} = \frac{(p_1-p_2)-(\pi_1-\pi_2)}{\sqrt{\bar{p}(1-\bar{p})(\frac{1}{n_1}+\frac{1}{n_2})}}$		
5	Reject H_0 if $z_{stat} < z_\alpha$, do not reject H_0 if $z_{stat} \geq z_\alpha$	Reject H_0 if $z_{stat} < z_{\frac{\alpha}{2}}$ or if $z_{stat} > z_{1-\frac{\alpha}{2}}$, do not reject H_0 if $z_{\frac{\alpha}{2}} \leq z_{stat} \leq z_{1-\frac{\alpha}{2}}$	Reject H_0 if $z_{stat} > z_{1-\alpha}$, do not reject H_0 if $z_{stat} \leq z_{1-\alpha}$
6	Reject H_0: Since $z_{stat} < z_\alpha$, reject the H_0. There is sufficient evidence that π_1 is smaller than π_2 Do not reject H_0: Since $z_{stat} \geq z_\alpha$, do not reject the H_0. There is not sufficient evidence that there is a difference between π_1 and π_2 or that π_1 is smaller than π_2	Reject H_0: Since $z_{stat} < z_{\frac{\alpha}{2}}$ (of $z_{stat} > z_{1-\frac{\alpha}{2}}$), reject the H_0. There is sufficient evidence that π_1 differs from π_2 Do not reject H_0: Since $z_{\frac{\alpha}{2}} \leq z_{stat} \leq z_{1-\frac{\alpha}{2}}$, do not reject the H_0. There is not sufficient evidence that π_1 differs from π_2	Reject H_0: Since $z_{stat} > z_{1-\alpha}$, reject the H_0. There is sufficient evidence that π_1 is larger than π_2 Do not reject H_0: Since $z_{stat} \leq z_{1-\alpha}$, do not reject the H_0. There is not sufficient evidence that there is a difference between π_1 and π_2 or that π_1 is larger than π_2

3.2.2 p-value approach

	Left-sided test	**Two-sided test**	**Right-sided test**
1	H_0: $\pi_1 - \pi_2 = 0$ (no difference) H_1: $\pi_1 - \pi_2 < 0$ ($\pi_1 < \pi_2$)	H_0: $\pi_1 - \pi_2 = 0$ (no difference) H_1: $\pi_1 - \pi_2 \neq 0$ ($\pi_1 \neq \pi_2$)	H_0: $\pi_1 - \pi_2 = 0$ (no difference) H_1: $\pi_1 - \pi_2 > 0$ ($\pi_1 > \pi_2$)

2	α, n_1, n_2		
4	$\bar{p} = \frac{x_1+x_2}{n_1+n_2}$ $z_{stat} = \frac{(p_1-p_2)-(\pi_1-\pi_2)}{\sqrt{\bar{p}(1-\bar{p})(\frac{1}{n_1}+\frac{1}{n_2})}}$		
5	p-value = $P(Z < z_{stat})$	If $z_{stat} < 0$, p-value = $2P(Z < z_{stat})$ or if $z_{stat} > 0$, p-value = $2P(Z > z_{stat})$	p-value = $P(Z > z_{stat})$
6	Reject H_0: Since p-value $<$ α, reject the H_0. There is sufficient evidence that π_1 is smaller than π_2 Do not reject H_0: Since p-value $\geq$ α, do not reject the H_0. There is not sufficient evidence that there is a difference between π_1 and π_2 or that π_1 is smaller than π_2	Reject H_0: Since p-value $<$ α, reject the H_0. There is sufficient evidence that π_1 differs from π_2 Do not reject H_0: Since p-value $\geq$ α, do not reject the H_0. There is not sufficient evidence that π_1 differs from π_2	Reject H_0: Since p-value $<$ α, reject the H_0. There is sufficient evidence that π_1 is larger than π_2 Do not reject H_0: Since p-value $\geq$ α, do not reject the H_0. There is not sufficient evidence that there is a difference between π_1 and π_2 or that π_1 is larger than π_2

4 Hypothesis testing for the difference in population variances

Test the hypothesis about the ratio between σ_1^2 and σ_2^2.

Estimate $\frac{\sigma_1^2}{\sigma_2^2}$ with $\frac{s_1^2}{s_2^2}$

Let s_1^2 be the larger variance estimate and s_2^2 the smaller variance estimate. This ensures that $F_{stat} \geq 1$.

Test statistic:

$$F_{stat} = \frac{s_1^2}{s_2^2}$$

Degrees of freedom:

$$\text{df}_1 = n_1 - 1$$

$$\text{df}_2 = n_2 - 1$$

Critical values:

Determine critical values $F_\alpha^{\text{df}_1,\text{df}_2}$ from F table. There is a table for each α. The tables available on EMSLearn have tables for $\alpha = 0.05$, $\alpha = 0.025$ and $\alpha = 0.01$.

Search for df_1 in rows of table and df_2 in columns.

4.1 Confidence interval

The confidence interval for $\frac{\sigma_1^2}{\sigma_2^2}$ will only be discussed in Statistics 214.

4.2 Hypothesis tests

Since it is ensured that $F_{stat} \geq 1$, there is not a left-sided test.

Firstly, determine which population is 1 and 2 by determining which sample variance is larger.

4.2.1 Critical value approach

	Two-sided test	**Right-sided test**
1	H_0: $\sigma_1^2 = \sigma_2^2$ vs H_0: $\sigma_1^2 \neq \sigma_2^2$	H_0: $\sigma_1^2 = \sigma_2^2$ vs H_0: $\sigma_1^2 > \sigma_2^2$
2	α, n_1, n_2	
3	$F_{\frac{\alpha}{2}}^{\mathsf{df}_1,\mathsf{df}_2}$	$F_{\alpha}^{\mathsf{df}_1,\mathsf{df}_2}$
	$\mathsf{df}_1 = n_1 - 1$ en $\mathsf{df}_2 = n_2 - 1$	
	Find F_{crit} value in table where $\frac{\alpha}{2}$ F.INV.RT($\frac{\alpha}{2}$,df_1,df_2)	Find F_{crit} value in table where α F.INV.RT(α,df_1,df_2)
4	$F_{stat} = \frac{s_1^2}{s_2^2}$	
5	Reject H_0 as $F_{stat} > F_{\frac{\alpha}{2}}^{\mathsf{df}_1,\mathsf{df}_2}$. Do not reject H_0 if $F_{stat} \geq F_{\frac{\alpha}{2}}^{\mathsf{df}_1,\mathsf{df}_2}$	Reject H_0 if $F_{stat} > F_{\alpha}^{\mathsf{df}_1,\mathsf{df}_2}$. Do not reject H_0 if $F_{stat} \geq F_{\alpha}^{\mathsf{df}_1,\mathsf{df}_2}$
6	Reject H_0: Since $F_{stat} > F_{\frac{\alpha}{2}}^{\mathsf{df}_1,\mathsf{df}_2}$, reject the H_0. There is sufficient evidence of a statistically significant difference between σ_1^2 and σ_2^2. Do not reject H_0: Since $F_{stat} \leq F_{\frac{\alpha}{2}}^{\mathsf{df}_1,\mathsf{df}_2}$, do not reject the H_0. There is not sufficient evidence of a statistically significant difference between σ_1^2 and σ_2^2.	Reject H_0: Since $F_{stat} > F_{\alpha}^{\mathsf{df}_1,\mathsf{df}_2}$, reject the H_0. There is sufficient evidence that σ_1^2 is larger than σ_2^2. Do not reject H_0: Since $F_{stat} \leq F_{\alpha}^{\mathsf{df}_1,\mathsf{df}_2}$, do not reject the H_0. There is not sufficient evidence that σ_1^2 is larger than σ_2^2.

4.2.2 p-value approach

Although the F distribution is not symmetric, the lower tail of the distribution is still taken into account in the two-tailed test: If we consider that $F_{\text{stat}} < 1$, we can obtain the inverse of F_{stat} by $\frac{1}{F_{\text{stat}}}$.

$$\begin{aligned}\text{p-value} &= P(F < \frac{1}{F_{\text{stat}}}) + P(F > F_{\text{stat}}) \\ &= P(F > F_{\text{stat}}) + P(F > F_{\text{stat}}) \\ &= 2P(F > F_{\text{stat}})\end{aligned}$$

	Two-sided test	**Right-sided test**
1	H_0: $\sigma_1^2 = \sigma_2^2$ vs H_0: $\sigma_1^2 \neq \sigma_2^2$	H_0: $\sigma_1^2 = \sigma_2^2$ vs H_0: $\sigma_1^2 > \sigma_2^2$
2	α, n_1, n_2	
3	$F_{stat} = \frac{s_1^2}{s_2^2}$	
4	$\text{df}_1 = n_1 - 1$ and $\text{df}_2 = n_2 - 1$	
	p-value = $2P(F > F_{stat})$ 2*F.DIST.RT(F_{stat}, df_1, df_2)	p-value = $P(F > F_{stat})$ F.DIST.RT(F_{stat}, df_1, df_2)
5	Reject H_0 if p-value $< \alpha$. Do not reject H_0 if p-value $\geq \alpha$	
6	Reject H_0: Since p-value $< \alpha$, reject the H_0.There is sufficient evidence of a statistically significant difference between σ_1^2 and σ_2^2. Do not reject H_0: Since p-value $\geq \alpha$, do not reject the H_0. There is not sufficient evidence of a statistically significant difference between σ_1^2 and σ_2^2.	Reject H_0: Since p-value $< \alpha$, reject the H_0. There is sufficient evidence that σ_1^2 is larger than σ_2^2. Do not reject H_0: Since p-value $\geq \alpha$, do not reject the H_0. There is not sufficient evidence that σ_1^2 is larger than σ_2^2.

5 Effect sizes

As the sample size increases, the result of the test becomes less significant. A small p-value for a test with a larger sample size is less significant than a large p-value for a test with a smaller sample size.

Goal: Find a way to determine the strength of a result $\longrightarrow$ effect size.

Determine effect size with formula:

$$ES_B = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2 + s_2^2}{2}}}$$

Or the width of the confidence interval

$ES_B \approx 0.2$	Small effect size
$ES_B \approx 0.5$	Medium effect size
$ES_B \geq 0.8$	Large effect size

6 Questions

6.1 Question 1

Nine experts have rated two coffee brands in taste experiment. A rating is given on a 7-point scale (1 = extremely pleasant) for four characteristics: taste, aroma, richness and acidity. The data contains the combined ratings for all four characteristics.

	Brand	
Expert	**A**	**B**
C.C.	24	26
S.E.	27	27
E.G.	19	22
B.I.	24	27
C.M.	22	25
C.N.	26	27
G.N.	27	26
R.M.	25	27
P.V.	22	23

1. Is there any evidence of a difference in the average ratings between the two brands? Use a 5% significance level.

2. Calculate the p-value and make a decision.

3. Construct and interpret a 95% confidence for the difference in average ratings between the two brands.

6.2 Question 2

Regard the survey results of two types of marketers. The first type mainly focuses on businesses (B2B) and the second type focuses mainly on consumers (B2C). There was reported that 515 B2B and 236 B2C marketers on average uses a social media tool. Suppose that the survey is based on 590 B2B marketers and 411 B2C marketers.

At a 5% significance level, is there any evidence of a difference between B2B marketers and B2C marketers according to the proportion that uses a social media tool?

6.3 Question 3

Given the following measures from two independent populations:

	Group 1	**Group 2**
Sample size	40	40
Sample mean	2980	3205
Sample standard deviation	1140	963
Population mean	2500	3000

For the alternative hypothesis H_1: $\mu_1 < \mu_2$, calculate the power of the test.